1.7 The three-dimensional differential wave equation

- Start from $$\nabla^2\psi = \frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

  $$\psi(x,y,z,t) = Ae^{i\left(\vec{k}\cdot\vec{r}\mp\omega t+\varepsilon\right)} = Ae^{i\left(k_x x+k_y y+k_z z\mp\omega t+\varepsilon\right)}$$

  Thus, $$\frac{\partial\psi}{\partial x} = ik_x\psi \Rightarrow \frac{\partial^2\psi}{\partial x^2} = -{k_x}^2\psi$$

  Similarly, $$\nabla^2\psi = -\left({k_x}^2 + {k_y}^2 + {k_z}^2\right)\psi = -k^2\psi$$

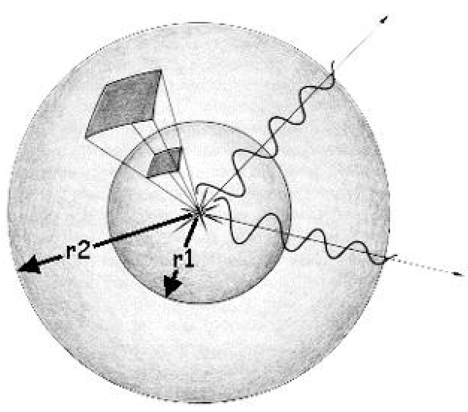
- Time derivative are given by

  $$\frac{\partial\psi}{\partial t} = -i\omega\psi \Rightarrow \frac{\partial^2\psi}{\partial t^2} = -\omega^2\psi$$

- Substitute (a) and (b) to wave equation: $$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

1.8 Spherical waves

- An idealized point source of light:
- The radiation emanating from it streams out radially, uniformly in all direction.
- Wavefronts: concentric spheres that increase in diameter as they expand out into the surrounding space.



Spherical waves

Sun light

- In spherical coordinates:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$

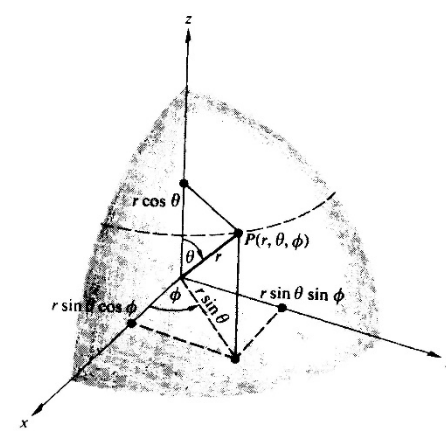
where $x = r sin\theta cos\varphi$, $y = r sin\theta sin\varphi$, $z = r cos\theta$

- Because waves are spherically symmetrical

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)$$

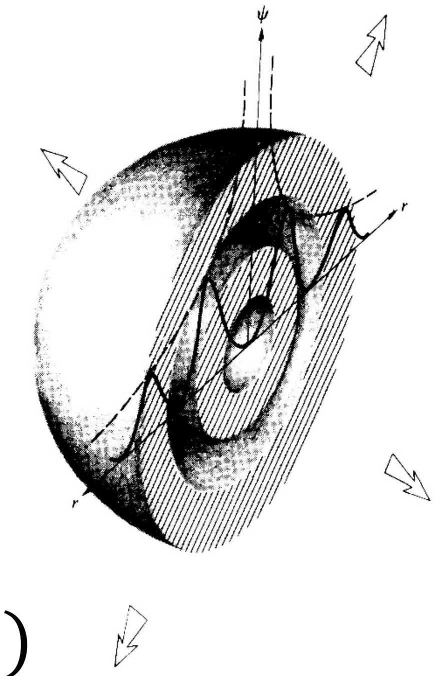
- The spherically symmetric differential wave

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,t)}{\partial r}\right) = \frac{1}{v^2}\frac{\partial^2\psi(r,t)}{\partial t^2}$$



- We have $\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{v^2}\frac{\partial^2(r\psi)}{\partial t^2}$
- The solution $r\psi(r,t) = f(r - vt)$
- Another solution $r\psi(r,t) = g(r + vt)$
- A special case of the general solution

$$\psi(r,t) = C_1\frac{f(r - vt)}{r} + C_1\frac{g(r + vt)}{r}$$



- The harmonic spherical wave

$$\psi(r,t) = \left(\frac{\mathcal{A}}{r}\right)\cos k(r \mp vt)$$

- Note: a spherical wave decreases in amplitudes as it expands from the origin.

Homework

Please drive $\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{v^2}\frac{\partial^2(r\psi)}{\partial t^2}$ from the spherically symmetric differential wave equation: $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,t)}{\partial r}\right) = \frac{1}{v^2}\frac{\partial^2\psi(r,t)}{\partial t^2}$